JEE Mains 2019 Chapter wise Question Bank

Waves and Sound - Questions

Q1

Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio:

(1) 16 : 9 (2) 25 : 9
(3) 4 : 1 (4) 5 : 3

9 Jan Morning

Q2

A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m (m<<M). When the car is at rest, the speeed of transverse waves in the string is 60 ms^{-1}. when the car has acceleration a, the wave-speed increases to 60.5 ms^{-1}. The value of a, in terms of gravitational acceleration g, is closest to:

(1) $\frac{g}{30}$ (2) $\frac{g}{5}$
(3) $\frac{g}{10}$ (4) $\frac{g}{20}$

9 Jan Morning

Q3

A musician using an open flute of length 50 cm produces second harmonic sound waves. A person runs towards the musician from another end of a hall at a speed of 10 km/h. If the wave speed is 330 m/s, the frequency heard by the running person shall be close to:

(1) 666 Hz (2) 753 Hz
(3) 500 Hz (4) 333 Hz

9 Jan Evening

Q4

A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes on the string is close to:

(1) 10.0 cm (2) 33.3 cm
(3) 16.6 cm (4) 20.0 cm

10 Jan Morning

Q5

A train moves towards a stationary observer with speed 34 m/s. The train sounds a whistle and its frequency registered by the observer is f_1 If the speed of the train is reduced to 17 m/s, the frequency registered is f_2 If speed of sound is 340 m/s, then the ratio f_1/f_2 is:

(1) 18/17 (2) 19/18
(3) 20/19 (4) 21/20

10 Jan Morning

Q6

A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be: (Assume that the highest frequency a person can hear is 20,000 Hz)

(1) 6 (2) 4
(3) 7 (4) 5

10 Jan Evening

Q7

Equation of travelling wave on a stretched string of linear density 5 g/m is y = 0.03 sin (450 t – 9x) where distance and time are measured in SI units. The tension in the string is:

(1) 10 N (2) 7.5 N
(3) 12.5 N (4) 5 N

11 Jan Morning

Q8

A travelling harmonic wave is represented by the equation $y(x, t)=10^{-3}\sin(50t+2x)$, where x and y are in meter and t is in seconds. Which of the following is a correct statement about the wave?

(1) The wave is propagating along the negative x-axis with speed 25 ms^{-1}.
(2) The wave is propagating along the positive x-axis with speed 100 ms^{-1}.
(3) The wave is propagating along the positive x-axis with speed 25 ms^{-1}.
(4) The wave is propagating along the negative x-axis with speed 100 ms^{-1}.

12 Jan Morning

Q9

A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60º with ground level. But he finds the aeroplane right vertically above his position. If v is the speed of sound, speed of the plane is:

(1) $\frac{\sqrt{3}}{2}v$ (2) $\frac{2v}{\sqrt{3}}$
(3) v (4) $\frac{v}{2}$

12 Jan Morning

Q10

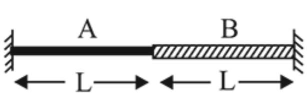
A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to:

(1) 322 ms^{-1} (2) 341 ms^{-1}
(3) 335 ms^{-1} (4) 328 ms^{-1}

12 Jan Evening

Q11

A wire of length 2L, is made by joining two wires A and B of same length but different radii r and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio $p : q$ is :



(1) 3 : 5 (2) 4 : 9 (3) 1 : 2 (4) 1 : 4

8 April Morning

Q12

The pressure wave, $P = 0.01\sin[1000t - 3x]$ Nm^{-2}, corresponds to the sound produced by a vibrating blade on a day when atmospheric temperature is 0°C. On some other day when temperature is T, the speed of sound produced by the same blade and at the same frequency is found to be 336 ms^{-1}. Approximate value of T is :

(1) 4°C (2) 11°C (3) 12°C (4) 15°C

9 April Morning

Q13

A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is $Y = 0.3\sin(0.157x)\cos(200\pi t)$. The length of the string is: (All quantities are in SI units.)

(1) 20 m (2) 80 m (3) 40 m (4) 60 m

9 April Morning

Q14

Two cars A and B are moving away from each other in opposite directions. Both the cars are moving with a speed of 20 ms^{-1} with respect to the ground. If an observer in car A detects a frequency 2000 Hz of the sound coming from car B, what is the natural frequency of the sound source in car B?

(speed of sound in air = 340 ms^{-1})

(1) 2250 Hz (2) 2060 Hz
(3) 2300 Hz (4) 2150 Hz

9 April Evening

Q15

A string 2.0 m long and fixed at its ends is driven by a 240 Hz vibrator. The string vibrates in its third harmonic mode. The speed of the wave and its fundamental frequency is:

(1) 180 m/s, 80 Hz (2) 320 m/s, 80 Hz
(3) 320 m/s, 120 Hz (4) 180 m/s, 120 Hz

9 April Evening

Q16

A stationary source emits sounds waves of frequency 500 Hz. Two observers moving along a line passing through the source detect sound to be of frequencies 4801 Hz and 530 Hz. Their respective speeds are, in ms^{-1},

(Given speed of sound = 300 m/s)

(1) 12, 16 (2) 12, 18 (3) 16, 14 (4) 8, 18

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Q17

The correct figure that shows, schematically, the wave pattern produced by superposition of two waves of frequencies 9 Hz and 11 Hz is :

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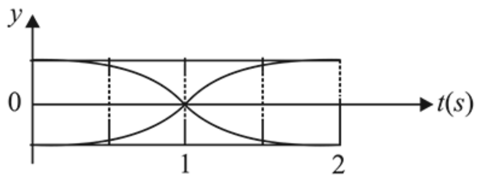

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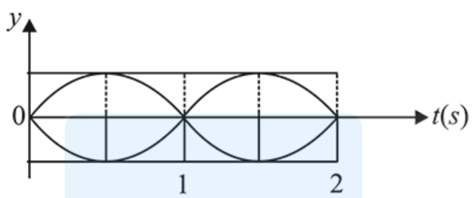

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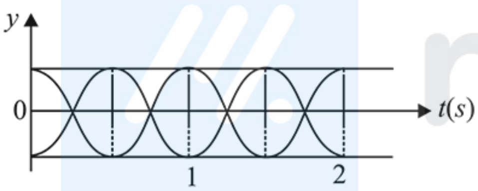

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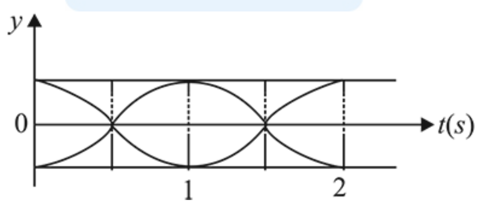

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Q18

A source of sound S is moving with a velocity of 50 m/s towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? (Take velocity of sound in air 350 m/s)

(1) 750 Hz (2) 857 Hz (3) 1143 Hz (4) 807 Hz

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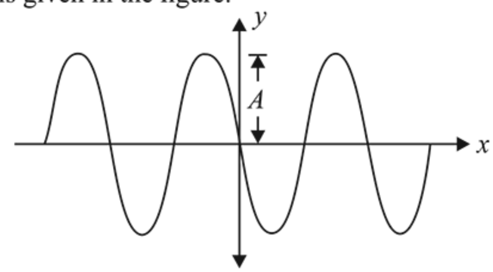
Q19

A progressive wave travelling along the positive x-direction is represented by $y(x,t) = A\sin(kx - \omega t + \phi)$. Its snapshot at $t = 0$ is given in the figure.



For this wave, the phase ϕ is :

(1) $-\frac{\pi}{2}$ (2) π (3) 0 (4) $\frac{\neq}{2}$

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Q20

A submarine (A) travelling at 18 km/hr is being chased along the line of its velocity by another submarine (B) travelling at 27 km/hr. B sends a sonar signal of 500 Hz to detect A and receives a reflected sound of frequency v. The value of v is close to :

(Speed of sound in water = 1500 ms^{-1})

(1) 504 Hz (2) 507 Hz
(3) 499 Hz (4) 502 Hz

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Q21

Two sources of sound S_1 and S_2 produce sound waves of same frequency 660 Hz. A listener is moving from source S_1 towards S_2 with a constant speed u m/s and he hears 10 beats/s. The velocity of sound is 330 m/s. Then u equals:

(1) 5.5 m/s (2) 15.0 m/s
(3) 2.5 m/s (4) 10.0 m/s

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Q22

A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound (v) in air by resonance tube method. Resonance is observed to occur at two successive lengths of the air column, $l_1 = 30$ cm and $l_2 = 70$ cm. Then, v is equal to :

(1) 332 ms^{-1} (2) 384 ms^{-1}
(3) 338 ms^{-1} (4) 379 ms^{-1}

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Q23

A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120 dB intensity sound ? [Given reference intensity of sound as 10^{-12}W/m^2]
(1) 40 cm (2) 20 cm (3) 10 cm (2) 30 cm

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Waves and Sound - Answers

Q1

(2) As we know, $\frac{I_{max}}{I_{min}} = \left(\frac{A_1+A_2}{A_1-A_2}\right)^2$

and $\sqrt{\frac{I_1}{I_2}} = \frac{A_1}{A_2}$

$\frac{I_{max}}{I_{min}} = 16 \Rightarrow \frac{A_{max}}{A_{min}} = 4 \Rightarrow \frac{A_1+A_2}{A_1-A_2} = \frac{4}{1}$

Using componendo and dividendo.

$\frac{A_1}{A_2} = \frac{5}{3} \Rightarrow \frac{I_1}{I_2} = \left(\frac{5}{3}\right)^2 = \frac{25}{9}$

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Q2

(2) Wave speed $V=\sqrt{\frac{T}{\mu}}$

when car is at rest $a = 0$

$\therefore 60 = \sqrt{\frac{Mg}{\mu}}$

Similarly when the car is moving with acceleration a,

$60.5 = \sqrt{\frac{M\left(g^2+a^2\right)^{1/2}}{\mu}}$

$\frac{60.5}{60} = \sqrt{\sqrt{\frac{g^2+a^2}{g^2}}}$

$\left(1+\frac{0.5}{60}\right)^4 = \frac{g^2+a^2}{g^2} = 1+\frac{2}{60}$

$\Rightarrow g^2+a^2 = g^2+g^2\times\frac{2}{60}$

$a = g\sqrt{\frac{2}{60}} = \frac{g}{\sqrt{30}}$

which is closets to g/5.

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Q3

(1) Frequency of the sound produced by open flute.

$f = 2\left(\frac{v}{2\ell}\right) = \frac{2\times330}{2\times0.5} = 660Hz$

Velocity of observer, $v_0 = 10\times\frac{5}{18} = \frac{25}{9} m/s$

As the source is moving towards the observer therefore, according to Doppler's effect.

$\therefore$ Frequency detected by observer,

$f' = \left[\frac{v+v_0}{v}\right] f = \left[\frac{\frac{25}{9}+330}{330}\right] 660$

$= \frac{2995}{9\times330}\times660$ or, $f' = 665.55 \simeq 666$ Hz

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Q4

(4) Velocity of wave on string

$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{8}{5}\times1000} = 40 m/s$

Here, T = tension and μ = mass/length

Wavelength of wave $\lambda = \frac{v}{n} = \frac{40}{100} m$

Separation b/w successive nodes,

$\frac{\lambda}{2} = \frac{40}{2\times100} = \frac{20}{100} m = 20$ cm

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Q5

(2) According to Doppler's effect, when source is moving but observer at rest

$$f_{app} = f_0\left[\frac{V}{V-V_s}\right] \Rightarrow f_1 = f_0\left[\frac{340}{340-34}\right]$$

and, $f_2 = f_0\left[\frac{340}{340-17}\right]$

$$\therefore \frac{f_1}{f_2} = \frac{340-17}{340-34} = \frac{323}{306} \text{ or, } \frac{f_1}{f_2} = \frac{19}{18}$$

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Q6

(1) If a closed pipe vibration in Nth mode then frequency of vibration $n = \frac{(2N-1)v}{4l} = (2N-1)n_1$

(where n_1 = fundamental frequency of vibration)

Hence $20{,}000 = (2N-1) \times 1500$

$\Rightarrow N = 7.1 \approx 7$

$\therefore$ Number of over tones = (No. of mode of vibration) – 1 $= 7 - 1 = 6$

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Q7

(3) We have given,

$y = 0.03 \sin(450t - 9x)$

Comparing it with standard equation of wave, we get

$\omega = 450 \; k = 9$

$$\therefore v = \frac{\omega}{k} = \frac{450}{9} = 50 \text{ m/s}$$

Velocity of travelling wave on a stretched string is given by

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{T}{\mu} = 2500$$

μ = linear mass density

$\Rightarrow T = 2500 \times 5 \times 10^{-3}$

$\Rightarrow 12.5$ N

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Q8

(1) Comparing the given equation

$y = 10^{-3}\sin(50t + 2x)$ with standard equation,

$y = a\sin(\omega t - kx)$

$\Rightarrow$ wave is moving along –ve x-axis with speed

$$v = \frac{\omega}{K} \Rightarrow v = \frac{50}{2} = 25 \text{ m/sec.}$$

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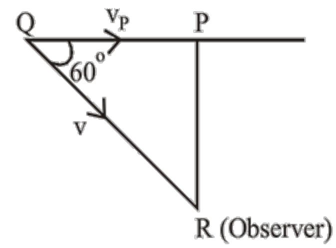
Q9

(4)



Distance, $PQ = v_p \times t$ (Distance = speed × time)

Distance, $QR = V.t$

$$\cos 60° = \frac{PQ}{QR}$$

$$\frac{1}{2} = \frac{v_p \times t}{V.t} \Rightarrow v_p = \frac{v}{2}$$

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Q10

(4)

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Q11

(3) As there must be node at both ends and at the joint of the wire A and B so

$$\frac{V_A}{V_B} = \sqrt{\frac{u_B}{u_A}} = \frac{r_B}{r_A} = 2 = \frac{\lambda_A}{\lambda_B}$$

$$\Rightarrow \lambda_A = 2\lambda_B \Rightarrow \frac{P}{q} = \frac{1}{2}$$

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Q12

(1) On comparing with $P = P_0 \sin(\omega t - kx)$, we have $\omega = 1000$ rad/s, $K = 3\ m^{-1}$

$$\therefore v_0 = \frac{\omega}{k} = \frac{1000}{3} = 333.3 \text{m/s}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\text{or, } \frac{333.3}{336} = \sqrt{\frac{273+0}{273+t}} \therefore t = 4°C$$

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Q13

(2) Given, $y = 0.3 \sin(0.157x)\cos(200\pi t)$

So, $k = 0.157$ and $\omega = 200\pi = 2\pi f$

$$\therefore f = 100 \text{ Hz and } v = \frac{\omega}{k} = \frac{200\pi}{0.157} = 4000 \text{m/s}$$

Now, using $f = \frac{nv}{2l} = \frac{4v}{2l} = \frac{2v}{l}$ [here n = 4]

$$\therefore l = \frac{2v}{f} = \frac{2\times 4000}{100} = 80m$$

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Q14

(1) $f' = f\dfrac{v - v_0}{v + v_s}$

$$\text{or } 2000 = f\frac{340-20}{340+20}$$

$$\therefore f = 2250 \text{ Hz.}$$

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Q15

(2) $\dfrac{3\lambda}{2} = 2$ or $\lambda = \dfrac{4}{3}m$

$$\text{Velocity, } v = f_\lambda = 240\times\frac{4}{3} = 320 \text{ m/sec}$$

$$\text{Also } f_1 = \frac{240}{3} = 80 \text{ Hz}$$

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Q16

(2) Frequency of sound source $(f_0) = 500$ Hz

When observer is moving away from the source

Apparent frequency $f_1 = 480 = f_0\left(\dfrac{v - v_0'}{v}\right)$ ....(i)

And when observer is moving towards the source

$$f_2 = 530 = f_0\left(\frac{v - v_0''}{v}\right) \quad ....(ii)$$

From equation (i)

$$480 = 500\left(\frac{300 - v_0'}{300}\right)$$

$v_0' = 12$ m/s

From equation (ii)

$$530 = 500\left(1 + \frac{v_0''}{v}\right)$$

$\therefore V_0'' = 18$ m/s

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Q17

(3) Beat frequency

= difference in frequencies of two waves

= 11 – 9 = 2 Hz

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Q18

(1) When source is moving towards a stationary observer,

$$f_{app} = f_{source}\left(\frac{V-0}{V-50}\right)$$

$$1000 = f_{source}\left(\frac{350}{300}\right)$$

When source is moving away from observer

$$f' = f_{source}\left(\frac{350}{350+50}\right)$$

$$f' = \frac{1000\times 300}{350}\times\frac{350}{400}$$

$$f' \approx 750 \text{ Hz}$$

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Q19

(2) At $t = 0, x = 0, y = 0$

$\therefore \quad \phi = \pi$ rad

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Q20

(4) $f_1 = f\left(\dfrac{v-v_o}{v-v_s}\right) = f\left(\dfrac{1500-5}{1500-7.5}\right)$

No reflected signal,

$f_2 = f_1\left(\dfrac{v+v_o}{v+v_s}\right) = f_1\left(\dfrac{1500+7.5}{1500+5}\right)$

$f_2 = 500\left(\dfrac{1500-5}{1500-7.5}\right)\left(\dfrac{1500+7.5}{1500+5}\right)$

$\simeq$ 502 Hz

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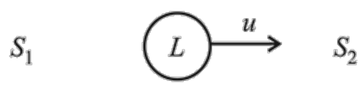
Q21

(3) $f_1 = f\dfrac{v-v_0}{v}$ and $f_2 = f\dfrac{v+v_0}{v}$



But frequency,

$f_2 - f_1 = f \times \dfrac{2v_0}{v}$

or $10 = 660 \times \dfrac{2u}{330}$

$\therefore$ $u = 2.5$ m/s.

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Q22

(2) $V = f\lambda = f \times 2(\ell_2 - \ell_1)$
$= 480 \times 2(0.70 - 0.30)$
$= 384$ m/s

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Q23

(1) Using, $\beta = 10 \log_{10}\left(\dfrac{I}{I_0}\right)$

or $120 = 10 \log 10\left(\dfrac{I}{10^{-12}}\right)$ ...(i)

Also $I = \dfrac{P}{4\pi r^2} = \dfrac{2}{4\pi r^2}$ ...(ii)

On solving above equations, we get
$r = 40$ cm.

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